



Musk wants a Pizza joint on Mars

While announcing plans to test fire the rocket that will take humans to Mars, Musk also indicated that Mars would need pizza joints <http://digit.in/marspizza>



Hawking urged humans to leave

He saw a bleak future for humanity, unless efforts were taken to establish a permanent settlement outside the Earth <http://digit.in/hawkingmars>

process. The raw materials required are water ice and carbon dioxide, both of which are available on Mars. A high temperature reaction with a catalyst can provide the required methane, along with water. The water can be reused as a raw material for the process. The same approach can also be used to produce rocket fuel on Earth.

FOOD

Pioneers and the eventual colonists on Mars will have to grow their own crops. NASA has used the Veggie system to grow crops on board the ISS. The system has been used to grow lettuce and cabbage. The system uses a combination of green, blue and red lights to provide the energy for the photosynthesis.

The challenges involved are identifying the specific varieties of crops that are ideal for growing in the low gravity environments. NASA is testing out a number of crops on Earth to see which ones are ideal for growth in space. Making sure these crops are safe to eat is another concern. Potatoes provide twice the amount of food as compared to seed crops, by using the same amount of light. Potatoes are a good source of carbohydrates, while soybeans are a good source of proteins. Along with wheat and salad crops, a well balanced diet can be provided by the greenhouses.

The benefits of the portable greenhouses do not end at just providing the food. They can play an important role in waste and water recycling as well as air revitalisation. The plants can convert the carbon dioxide exhaled by the astronauts into oxygen and food. The greenhouses can be autonomous.

WASTE DISPOSAL

NASA is working on a trash reactor that can convert waste into rocket fuel. The reactor burns the waste at 537 degrees Celsius, which produces methane, oxygen and water. Using this reactor, about 4.5 kg of waste can be converted to 3.17 kg of fuel.

Another experimental waste disposal device, known as the Vortical Oxidative Reactor Technology Experiment (VORTEX) also burns up the waste. The system can be used to dispose organic waste with ash, water, heat and carbon dioxide as the by-products. Two of the products, ash and carbon dioxide can be used by the plants, as fertiliser and for photosynthesis.

HABITATS

A colony or a pioneering mission to Mars would require the use of construction of structures to house the humans, the equipment, the greenhouses for the food, and other facilities. The simplest approach is

stone. Genetically engineered organisms can produce the protein in biological factories. It is then mixed with the locally available soil using vacuum technology.

The regolith on Mars has properties that make it ideal for making bricks. Research has shown that the iron oxide within the Martian soil acts as a binding agent. With minimal resources, it is possible to convert the Martian regolith into bricks, by just applying pressure.

Researchers have demonstrated the viability of using 3D printers not just for structures, but for tools as well. Simple solvents and biopolymers, combined with the simulants of Lunar and Martian soil,

have been used to create "inks" for 3D printing. These can then be used to create flexible and elastic structures, as well as bricks that fit together like LEGO blocks.

The rovers by NASA have identified silicon, calcium, chlorine, iron and titanium on Mars.

These can be used to make a variety of

materials for use by humans, including plastics, metals, glass and paper.

ISRO wants to get in on the action and compares setting up a base on the Moon to building a research outpost in Antarctica. ISRO wants to use the moon soil as the raw material for 3D printing Igloos. Towards this approach, ISRO has produced soil on Earth which mimics the properties of Lunar soil.

A permanent presence on the Moon can allow for testing of equipment and a launching platform for missions into the outer solar system. The manufacturing of some equipment, such as solar panels, as well as putting them into space, is theoretically easier on the Moon than it is on Earth. 

Credit: NASA



A concept of an inflatable greenhouse on Mars.

to use the locally available soil, or the regolith and convert it into the construction materials required for structures. The structures are then 3D printed by robots.

The structures on the Moon and Mars for human housing and facilities would require thousands of tonnes of concrete. This is to protect the humans, equipment, crops and provisions from a constant bombardment of radiation and micrometeorites.

Conventional approaches to making concrete is an energy hungry process. Additionally, cement mixers can not work in the low gravity environments.

A novel approach involves a new form of concrete known as bio-concrete. The binding agent used is an animal protein instead of lime-